- Utilized YOLO v5 to develop an object-detection model in PyTorch that achieved a 80% mAP evaluation on an in-house thermal-image vehicle dataset for a R&D project.
- Designed an anomaly-detection model in PyTorch using an LSTM variational-autoencoder to detect component failure in a pulse-power x-ray machine. The model accurately predicted the location of a component failure with 87% accuracy and cut machine repair times by 50%.
- Applied the U-Net architecture with PyTorch to develop a deblurring model for a pulse-powered x-ray machine. The model was applied by the machine operators and reduced machine configuration time by 30%.
- Enhanced the scalability of models initially prototyped by our team of Analysts and Data Scientists for training on our internal high-performance computing (HPC) cluster. My role involved refactoring the existing code to enable efficient execution in a distributed and parallel setting.
- Mentored a summer intern in the development and execution of models on a HPC cluster, involving multi-node, multi-GPU setups and Slurm resource management.
- Developed and deployed a Django web interface with Docker Compose, enabling non-technical staff to add nodes and connections to the company's Neo4j graph database, without knowledge of the query syntax.
- Developed interactive dashboards using Streamlit to give non-technical staff the abilty to interact with the capabilities designed by our team of Data Scientists.